

Youbin He

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EDUCATION

Shandong Jiaotong University, Jinan, China

Sep 2019 – Jun 2023

BS in Geographic Information Science, GPA 89.6 / 100 (Top 3%)

PUBLICATIONS

- [1] He, Y., Yang, Y. (2025). Research on the spatial distribution and influencing factors of B&Bs in Liangshan prefecture based on GIS spatial analysis and spatial econometric model. In: Proceedings of the International Conference on Mechatronics and Intelligent Control (ICMIC 2024). SPIE, 13447: 1300-1320.
- [2] Wang, R., He, Y., Li, S., et al. (2024). Forest fire monitoring based on remote sensing technology. In: Proceedings of the Fifth International Conference on Geology, Mapping, and Remote Sensing (ICGMRS 2024). SPIE, 13223: 505-508.
- [3] He, Y., Ju, X. (2024). Xichang City land dynamic changes based on remote sensing data. In: Proceedings of the Fourth International Conference on Geology, Mapping, and Remote Sensing (ICGMRS 2023). SPIE, 12978: 277-282.
- [4] He, Y., Zhou, S., Quan, X. (2022). Remote Sensing Image Scene Classification Based on ECA Attention Mechanism Convolutional Neural Network. In: Proceedings of the 2022 IEEE 4th International Conference on Civil Aviation Safety and Information Technology (ICCASIT). IEEE, 1265-1269.

RESEARCH EXPERIENCE

Shandong Jiaotong University, Jinan, China

Sep 2019 – Jun 2023

Multi-source Remote Sensing Data Applications for Ecological Environments - Research Team Leader

- **Land-Use Classification:** Applied supervised classification on preprocessed Landsat 8 time-series to delineate five land-use categories and quantified a 103.5 km² expansion of built-up areas.
- **Burn-Area Extraction:** Employed dNBR on Sentinel-2 and Landsat 8 imagery to estimate about 825 ha of burned area—within 4 % of the 791.6 ha field survey—enhancing fire-monitoring accuracy.
- **Ecological Degradation Assessment:** Leveraged the Remote Sensing Ecological Index (RSEI) to reveal post-fire ecosystem impacts: “excellent” ecological zones declined by 38.2 % and overall vegetation cover dropped by 3.3 %.
- **Key Achievements:** One first-author publication; One second-author publication; Awarded First Prize, Shandong Jiaotong University Remote Sensing Technology Competition.

Multi-source Data Applications for Urban Development - Research Team Leader

- **Designed Composite Assessment Model:** Constructed an AHP-based urban evaluation framework with six economic, ecological, and social-welfare indicators, partitioning development into three tiers for objective policy guidance.
- **Developed Multi-Scale Driver Model:** Integrated multi-source datasets (POI, OSM, NPP-VIIRS, etc.) and applied the Geographical Detector and Geographically Weighted Regression (GWR) to quantify drivers of B&B spatial distribution at both global and local scales in Liangshan.
- **Spatial Pattern Analysis:** Employed Average Nearest Neighbor Index and Standard Deviational Ellipse to uncover a NE–SW clustering of homestays centered on Xichang and Yanyuan.
- **Key Achievements:** One first-author publication; Awarded National Third Prize (×2), Esri China Cup National GIS Competition; Awarded First Prize (×2), Shandong Jiaotong University Remote Sensing

Technology Competition; Outstanding Undergraduate Thesis Award.

Intelligent Geospatial Scene Interpretation - Core Member

- **Model Development & Optimization:** Integrated Efficient Channel Attention (ECA) blocks into ResNet-34 residual modules to preserve feature dimensions; validated on the AID dataset, achieving 94.95 % accuracy—a 3.45 % improvement over the standard ResNet-34.
- **Key Achievements:** One first-author publication.

Capital Normal University, State Key Laboratory Cultivation Base of Urban Environmental Processes and Digital Simulation, Beijing, China

Jul 2022

2022 Summer Academic Camp for Outstanding Undergraduates - Distinguished Camp Participant

- **Academic Training & Presentation:** Participated in expert-led lectures on InSAR-based subsidence monitoring and sub-pixel mangrove estimation; presented undergraduate research findings; awarded Distinguished Camp Participant (Top 11%, 18 of 163).

WORK EXPERIENCE

XingTuZhiHui (Xi'an) Digital Technology Co.Ltd, Chengdu, China

May 2024 – Present

Remote Sensing Algorithm Engineer and Project Manager

- **Independent Algorithm Development:** Developed Python-based preprocessing algorithms (radiometric calibration, atmospheric & geometric corrections, image fusion) and automated batch L1 to L2 product generation for GaoFen-series (GF-1/2/6) and AQUA satellites.
- **Fire Monitoring Subsystem Development:** Assisted in MODIS workflow development for daily surface-temperature inversion, time-series compositing, and thematic-map production; supported B/S module development and testing with Piesat Information Technology Co., Ltd.
- **Agri-Monitoring Subsystem Development:** Assisted in developing automated workflows for FY-3D/E/F and FY-4B satellites to invert vegetation indices and produce crop-growth assessment products; independently developed a crop-yield estimation module for production forecasting.

AWARDS & HONORS

National Third Prize ×2, Esri-Cup Chinese College Students GIS Software Development Contest	2022 - 2023
Honorable Mention ×2, Esri-Cup Chinese College Students GIS Software Development Contest	2020 - 2021
First Prize ×3, Shandong Jiaotong University Remote Sensing Technology Competition	2021 - 2023
National Encouragement Scholarship×3	2019 - 2022
First-Class Scholarship ×3	2019 - 2022
Outstanding Graduate of Shandong Province	2023
Outstanding Student of Shandong Province	2022
Outstanding Undergraduate Thesis Award	2023

SKILLS

Programming Languages: Python(NumPy, GDAL, Rasterio, scikit-learn, PyTorch), SQL, C, C#, Matlab

Geospatial tools: GEE(JavaScript), ArcGIS Pro(ArcPy), ENVI, QGIS

Languages: IELTS (XX) RXX LXX SXX WXX; Mandarin(Native)